

Fernando Moreira da Silva

Full-Stack TypeScript (React/Next.js) · Application Security (OWASP) · Author of Nemesis Defender

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Portfolio: hub-fernando-dev.vercel.app · GitHub: github.com/feryamaha · LinkedIn: linkedin.com/in/feryamaha

PROFESSIONAL SUMMARY

Full-stack TypeScript developer (React/Next.js) focused on clean architecture, security by default, and AI-assisted development under deterministic governance (a custom multi-IDE harness). Author and maintainer of Nemesis Defender, an open-source security enforcement framework written in Rust with a kernel layer in eBPF/LSM on Linux. I apply OWASP, runtime validation, and supply-chain security in day-to-day production. I moved into software in 2022, after 17 years in industrial metrology, from which I bring explicit contracts, low tolerance for ambiguity, and verifiable quality. Currently pursuing a Software Engineering degree.

TECHNICAL SKILLS

Languages: TypeScript (strict mode), JavaScript (ES6+), Rust, SQL, HTML5, CSS3

AI-Assisted Development & Agentic Coding: Claude Code, Codex, Cursor, Windsurf, Devin, Antigravity, OpenCode, Grok, Gemini; LLM-agent governance via a custom multi-IDE harness: SDD pipeline (spec, critical analysis, plan, implementation, tests and doc-sync), deterministic pretool hooks (exit 2), subagent-driven development, trust ledger and versioned canonical rules

Frameworks & Libraries: React, Next.js (App Router, React Server Components), Node.js (BFF via Route Handlers), React Hook Form, Zod, Tailwind CSS, Recharts, next-intl (i18n), Framer Motion

Architecture: Clean Architecture, Design Systems, BFF (Backend for Frontend), layered separation (UI, Hooks, Services, Types), explicit TypeScript contracts

APIs & Data: REST API integration and aggregation with provider fallback, caching, rate limiting

Databases: PostgreSQL, Drizzle ORM (schema and migrations via drizzle-kit), Neon (serverless Postgres), data modeling

Security: OWASP, Content Security Policy (CSP Level 3, dynamic nonce), HSTS, X-Frame-Options, Permissions-Policy, authentication and session (Auth.js/NextAuth, bcrypt hashing), hardening, threat modeling, runtime validation

DevSecOps & Supply-Chain: working knowledge of SLSA (provenance attestation), CI audit gate (cargo audit), SHA-256 checksums, signed commits (DCO)

Systems & Low-Level: eBPF, BPF LSM (Linux kernel), POSIX contract (exit codes, stdio), Linux/macOS terminal, AST analysis (tree-sitter), static analysis, fail-closed design

Performance: RSC, SSR, SSG, ISR, bundle optimization, LCP/TTI metrics, Lighthouse

Testing & Quality: Playwright (E2E), ESLint and Biome (custom rules), strict typing, Zod validation, regression tests

DevOps & Tools: Git, GitHub Actions (CI), Cargo, Bun, npm, Yarn, Linux, Figma

PROJECTS

Nemesis Defender (release v0)

Author and maintainer · Rust + eBPF · AGPL-3.0 · github.com/feryamaha/Nemesis_Defender_v0

- Independently designed and implemented a three-layer security enforcement framework (pre-tool hooks, content scanner, and eBPF LSM in the Linux kernel) that blocks destructive commands and supply-chain malware before execution.
- Built over roughly 1.5 years inside real production projects and open-sourced after validating the thesis; during that period, no AI agent executed a destructive command or accessed a sensitive file without permission.
- Implemented a 6-stage scanning pipeline (AST via tree-sitter, byte-level, regex, deny-list, entropy, and recursive decoder), combining a curated deny-list and semantic-analysis visitors against prompt injection, decode-then-exec, credential exfiltration, reverse shells, and typosquatting.
- Developed the kernel layer in eBPF/BPF LSM (bprm_check_security and egress control via socket_connect) as an independent containment net for destructive commands, with local logging to a violations ledger.
- Validated the system with adversarial red-teaming using real LLM agents and a custom suite of 184 tests across 26 modules, including a module dedicated to measuring false positives; every bypass found was fixed and covered by regression tests.
- Structured the release pipeline with SHA-256 checksums, provenance attestation (SLSA), and a CI audit gate (pentest and cargo audit) before each release, with blocking rules embedded in the binary (tamper-proof).
- Applied fail-closed design, corroboration-model quarantine, and IDE-agnostic architecture (Claude Code, Codex, Cursor, VS Code, Devin), validated on Linux and macOS, with complete technical documentation under AGPL-3.0.

PROFESSIONAL EXPERIENCE

Full-Stack Developer (TypeScript)

Nov 2024 - Present

- Developed the frontend of multi-role portals (5 access profiles), dashboards, and hotspots for the dental sector, with Next.js (App Router), React, and TypeScript
- Implemented a BFF via Route Handlers, keeping credentials and sensitive rules exclusively on the server.
- Structured the frontend architecture in layers (UI, Hooks, Services, Types) with explicit contracts and strict separation of responsibilities.
- Applied security by default aligned with OWASP: CSP Level 3 with dynamic nonce, HSTS, X-Frame-Options, and runtime validation with Zod.
- Built and maintained a Design System in Tailwind CSS with 90+ typed components and semantic tokens, reused across different client portals, accelerating feature delivery.
- Optimized rendering performance (RSC, SSR, SSG, ISR), keeping Lighthouse at 90+ across performance, accessibility, best practices, and SEO
- Led AI-assisted development (Devin/Windsurf, Claude, Codex, Cursor) under explicit architectural governance: features, refactors, fixes, and documentation.

Laboratory Technician / Metrologist

Jan 2007 - Nov 2024

Unipac / Síntegra Surgical Sciences · Pompeia, SP

- 17.5 years in dimensional metrology with advanced GD&T, technical-drawing interpretation, and reverse engineering, ensuring millimeter precision and quality compliance.
- Developed measurement programs for CMM (Mitutoyo, PolyWorks, PC-DMIS) and MSA and SPC studies, APQP Core Tools.
- A foundation of technical discipline that shapes how I build software: explicit contracts, low tolerance for ambiguity, and verifiable quality.

EDUCATION

Bachelor's in Software Engineering (in progress)

Jan 2026

Universidade Estácio de Sá

Self-Taught Program in Software Engineering

Nov 2022 - Present

Frontend Software Engineering — Architecture, Security, and AI Governance

Self-directed study following the Software Engineering curriculum, plus free online programming courses across various platforms. Intensive practice on real products: frontend architecture, critical analysis of complex systems, internal frameworks, design systems, applied security, and AI governance.

Technical Diploma in Plastics

2016 - 2018

SENAI Shunji Nishimura

CERTIFICATIONS & CONTINUING EDUCATION

- AI Governance Professional · Securiti.ai
- OWASP Top 10 for LLM Applications · OWASP Foundation
- AWS Cloud Fundamentals · AWS Skill Builder

LANGUAGES & AVAILABILITY

Languages: Portuguese (native) · English (technical reading; improving) · Spanish (intermediate)

Availability: Full-time (CLT), contractor (PJ), or freelance · Remote or hybrid